MASTER'S THESIS DEFENSE

Demand Forecasting and Inventory Optimisation Using Machine Learning

A simulation-based evaluation on the M5-Forecasting (Walmart) dataset, with an SME-accessible Power BI dashboard

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ROADMAP

What This Defense Covers

01

The Business Problem

Why forecasting alone doesn't solve inventory decisions

02

Methodology

Five model tiers, OUTL policy, and forward simulation

03

Results

Forecast accuracy and the central inventory finding

04

Recommendations

A diagnostic tool for SME decision-makers

Every Retailer Faces the Same Trade-off

Too little stock

→ lost sales, eroded customer trust

Too much stock

→ tied-up cash, holding cost, spoilage risk

Large retailers solve this with dedicated data-science teams and enterprise software. Small and medium-sized enterprises (SMEs) typically have neither — yet face an identical operational problem.

Published claims say:

10–25%

forecast accuracy improvement

5–15%

inventory cost reduction — from published ML forecasting studies

Source: Barghi (2025); Seyedan, Mafakheri & Wang (2023)

WHY THIS IS GENUINELY HARD

Four Structural Gaps in Current Practice

GAP 1

Forecast ≠ Decision

Most studies report accuracy and stop — they never translate forecasts into inventory outcomes.

GAP 2

No Live Validation

Inventory performance is usually assumed analytically, not measured by running the policy forward.

GAP 3

SMEs Left Behind

Enterprise tools assume infrastructure and staffing SMEs simply don't have.

GAP 4

Uncertainty Ignored

Textbook safety-stock formulas assume symmetric errors — real retail demand is skewed and intermittent.

RESEARCH OBJECTIVES

Five Objectives Guided This Research

1. Comparatively evaluate forecasting approaches — classical, ML, deep learning, ensemble, and probabilistic — on a hierarchical retail benchmark.

2. Develop a systematic feature-engineering strategy capturing the principal drivers of retail demand.

3. Integrate the best-performing forecast with the Order-Up-To-Level (OUTL) inventory policy.

4. Test whether literature-reported gains (10–25% accuracy, 5–15% cost) transfer to SKU-daily granularity.

5. Deliver a Power BI dashboard translating outputs into an accessible tool for SME decision-makers.

LITERATURE REVIEW — 52 SOURCES SYNTHESISED

A Consistent Finding, With One Critical Caveat

ML and ensemble methods consistently outperform classical statistics

across retail forecasting studies — this is well replicated.

Barghi (2025); Nasseri et al. (2023); Razzak et al. (2025) — 72-study systematic review

The caveat that matters most:
The published 10–25% / 5–15% benchmarks are measured on data aggregated to the weekly or category level — not the daily, individual-product level where real inventory decisions are made.

Benchmark source: Barghi (2025); Seyedan, Mafakheri & Wang (2023)

This thesis was designed to directly test whether those benchmark figures survive the move to SKU-daily granularity, using real product-store-day data and inventory outcomes measured — not assumed.

M5-Forecasting (Walmart) Dataset

3,049	**10**	**1,913**	**502**	**56**
products	stores, 3 US states	days (2011–2016)	series stratified subsample used	day out-of-sample test window

Chosen for scale, hierarchical structure mirroring real SME retail operations, and status as an established academic benchmark. A stratified subsample (proportional across category × state) was used for computational tractability.

Five Model Tiers, Compared Head-to-Head

Tier 1
Classical
MA, SES, AutoARIMA, Croston, Croston-SBA, Seasonal Naive

Tier 2
Standard ML
Random Forest, XGBoost, LightGBM (Tweedie loss)

Tier 3
Deep Learning
LSTM — 128/64 stacked units, dropout 0.2

Tier 4
Ensemble
Stacking: LGBM + XGB + RF → Ridge meta-learner

Tier 5
Probabilistic
LightGBM quantile regression (P10 – P99)

Trained strictly on historical data. Evaluated exclusively on a later, entirely unseen test period.

Croston / Croston-SBA for intermittent demand: Croston (1972); Syntetos & Boylan (2005).

METHODOLOGY

Translating Forecasts Into Inventory Decisions

The Order-Up-To-Level (R, s, S) Policy

A periodic review policy: every R days, if the inventory position falls at or below reorder point s, an order is placed to raise stock to order-up-to level S.

Three policy variants compared — differing only in how safety stock is calculated:

Classical
— Gaussian formula fed by classical forecast

ML-Gaussian
— Gaussian formula fed by ML ensemble forecast

ML-Empirical-Quantile
— distribution-aware formula from quantile model

(R, s, S) / OUTL framework: Silver, Pyke & Thomas (2017)

Safety Stock Formulas

$$SS = z \cdot \sigma \cdot \sqrt{(R + L)}$$

Gaussian (textbook) — assumes symmetric forecast-error distribution

$$SS = (P95 - P50) \cdot \sqrt{(R + L)}$$

Empirical quantile — reflects the true, skewed shape of retail demand

THE PIVOTAL METHODOLOGICAL DECISION

Measuring Performance, Not Assuming It

Most Published Studies

Calculate expected cost and service level analytically — using a formula to project what should happen under an assumed stockout rate.

This Thesis

Runs each policy forward, day by day, against real 56-day test demand for all 502 products — genuinely tracking stock, genuinely recording every stockout.

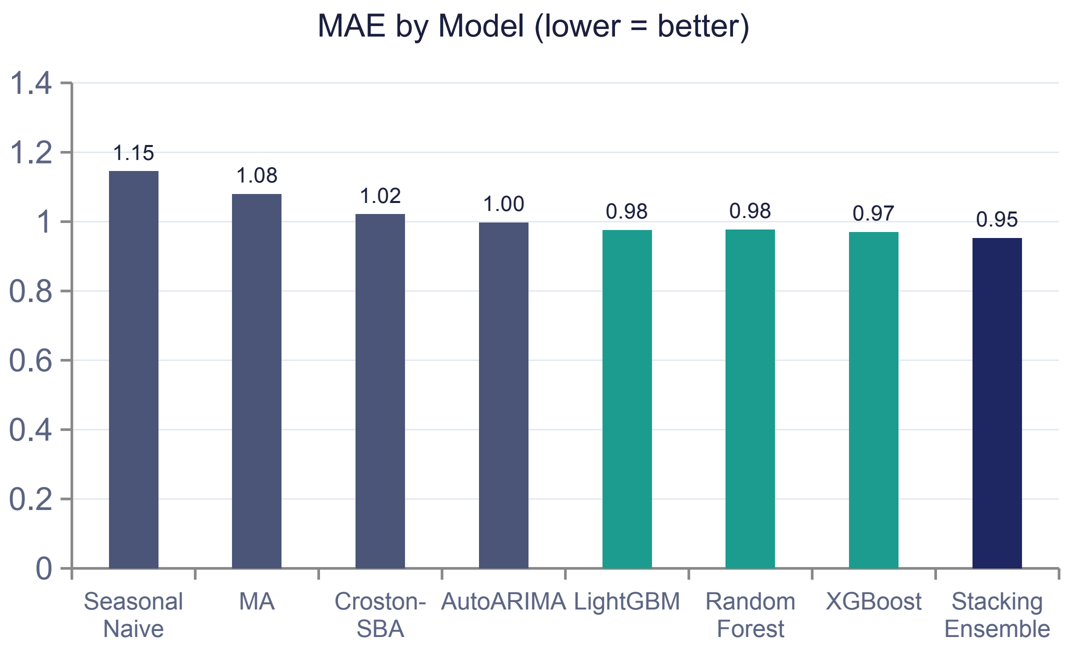
Falsification Test: Can It Detect a Stockout?

Before trusting the simulation, it was deliberately stress-tested under conditions engineered to force stockouts.

Stress Condition	Realised SL	Stockout Cost
Real buffers (baseline)	100.0%	$0
Safety stock set to zero	74.0%	$124,338
Demand scaled ×3	94.1%	$87,691
Replenishment suppressed	0.0%	$469,556

Every stress condition correctly registered degraded service and rising cost — confirming the 100% baseline result reflects demand genuinely being met, not a counter that never fires.

The Ensemble Wins, But the Margin Is Modest



+4.5%

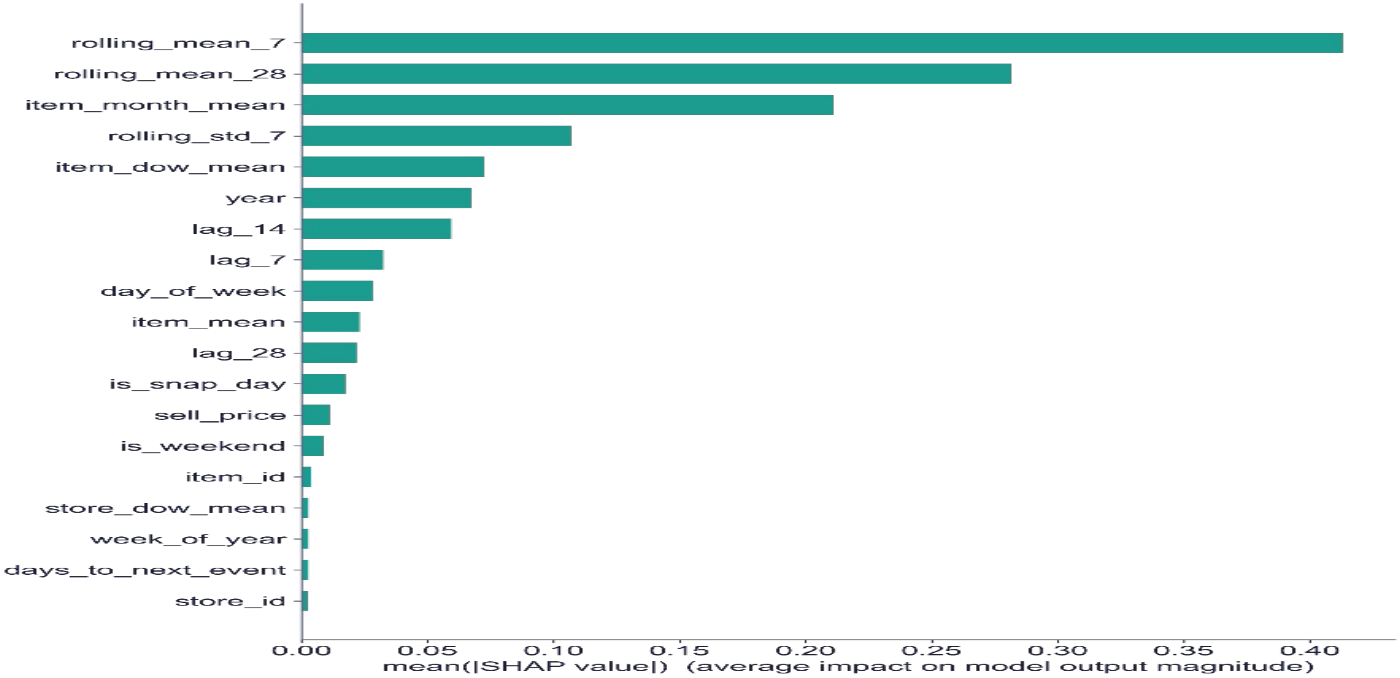
MAE improvement vs best classical baseline

+9.5%

RMSE improvement — large-error / spike sensitivity

MAPE improvement was essentially flat — below the 10–25% literature benchmark (Barghi, 2025; Seyedan et al., 2023), as anticipated (Slide 6).

What Actually Drives the Forecast



Top 5 features explain 78.6% of model behaviour — dominated by recent sales trend and item-level seasonality, not opaque signals.

THE CENTRAL FINDING

100%

Realised Service Level

— achieved by ALL THREE policies (Classical, ML-Gaussian, ML-Quantile) across every one of the 502 products, over the full 56-day test window. No policy ever stocked out.

Consequence:

With zero stockouts, there was no stockout cost left for a smarter forecast or a smarter safety-stock formula to save.

ML-Quantile policy cost:

+1.5%

more than the classical baseline — unnecessary safety stock, no offsetting benefit

AN HONEST CORRECTION

Why the Earlier 21% Estimate Did Not Survive

Earlier (Analytical) Draft

21%

Assumed the Gaussian policy achieved ~83% service and the quantile policy ~95% "by calibration," then computed cost from that gap.

The quantile policy won by construction — before any simulation ran.

This Thesis (Simulated)

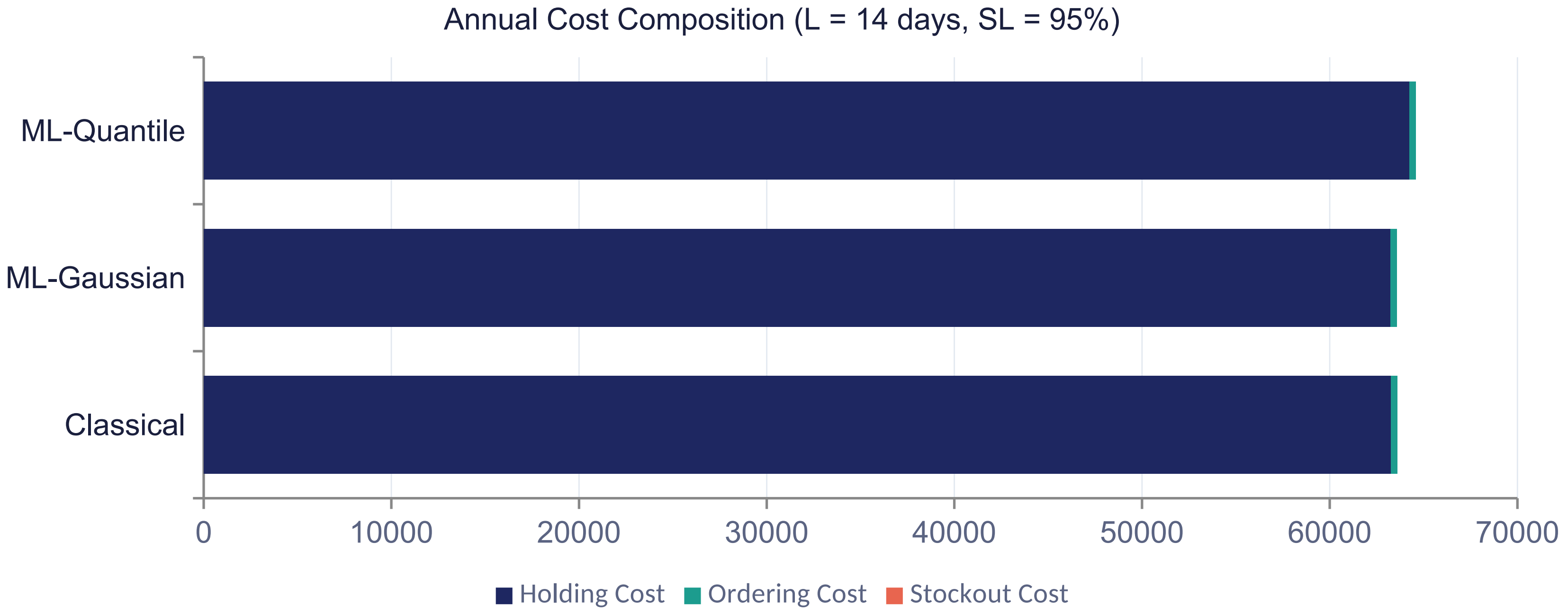
−1.5%

Both policies actually achieve 100% service — the quantile buffer protects against stockouts that never occur.

The Ch1 5–15% inventory-cost target is not achieved on this dataset — treated as a finding, not a failure.

5–15% target sourced from Barghi (2025); Seyedan, Mafakheri & Wang (2023)

Holding Cost Dominates; Stockout Cost Is Zero



Bars start at $0 — the ML-Quantile difference is a genuine but small ~1.5% increase, not the large gap a truncated axis would suggest.

The Finding Is Robust — Not a One-Off Result

Across Lead Time (7–21 days)

–1.3% to –1.7%

Quantile policy consistently more expensive, never favourable

Across Stockout Multiplier (0.4×–2.0×)

No change

Multiplier never engages — zero stockouts to price at any level

A threefold demand-shock stress test (Slide 11) shows even that is not enough, on this dataset, to make the quantile policy clearly superior.

THE REAL CONTRIBUTION

A Boundary Condition, Not a Blanket Claim

The value of ML forecasting and distributional safety stock is conditional on the demand regime:

Genuine stockout risk exists

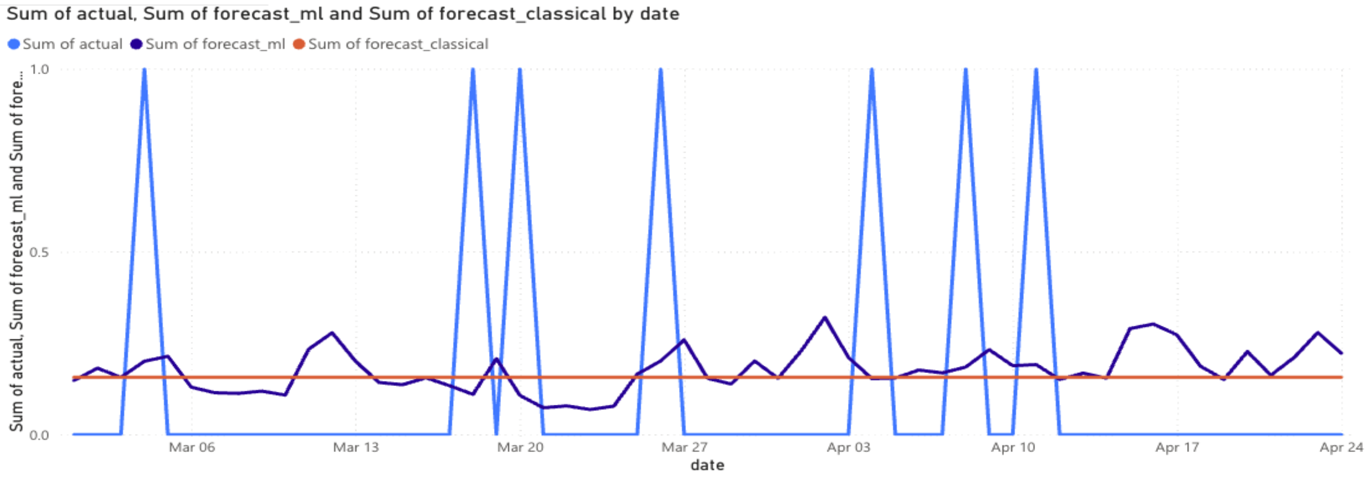
→ the larger, empirically calibrated buffer earns back its holding cost by preventing real stockouts.

Service is already saturated

→ the buffer protects against stockouts that don't occur — pure added cost, no benefit. (The M5 test regime.)

An Accessible Power BI Dashboard for SMEs

0.952	**100.0%**	**−1.5%**
MAE Ensemble	SL Quantile	Cost Chg. Quantile



Forecast Overview — actual vs. ML-ensemble vs. classical forecast, with KPI tiles and per-SKU drilldown

THE PRACTICAL DELIVERABLE

Cost Analysis — The Finding, Made Visible

Cost Analysis - Holding, Ordering, and Stockout Components

lead_time: 14

policy: ML_EMP_QUANTILE

Total Annual Cost	Holding Cost	Ordering Cost	Stockout Cost
64.57K	64.24K	331.82	0
Sum of total_cost	Sum of holding_cost	Sum of ordering_cost	Sum of stockout_cost

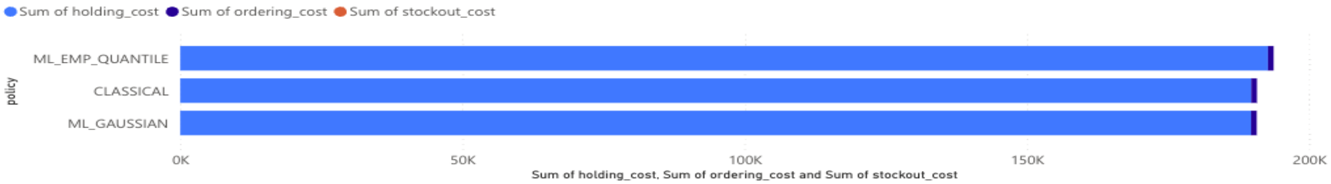
KPI tiles: ML-Empirical-Quantile policy, L = 14, m = 1.0 (central case, matches Ch5 Table 5.2)

policy	Average of realised_sl_pct	Sum of total_cost
CLASSICAL	100.00	758,748.24
ML_EMP_QUANTILE	100.00	770,189.76
ML_GAUSSIAN	100.00	758,605.71
Total	**100.00**	**2,287,543.71**

Total Annual Cost($) by Lead Time x Policy

lead_time	CLASSICAL	ML_EMP_QUANTILE	ML_GAUSSIAN	**Total**
7	183,060.51	185,473.89	183,030.45	**551,564.85**
10	186,374.55	189,033.93	186,341.43	**561,749.91**
14	190,762.50	193,718.25	190,725.66	**575,206.41**
21	198,550.68	201,963.69	198,508.17	**599,022.54**
Total	**758,748.24**	**770,189.76**	**758,605.71**	**2,287,543.71**

Cost Composition by Policy (L=14)



Free (Power BI free tier). Runs on a standard laptop. Requires only data a typical POS system already produces.

Theoretical and Practical Contributions

Theoretical

- First simulation-based measurement of the Gaussian-vs-empirical safety-stock choice on M5
- Demonstrates ML forecasting's inventory value is regime-conditional, not automatic
- Shows the inventory comparison is robust to the stockout-cost multiplier (zero stockouts at 0.4×–2.0×)

Practical

- End-to-end, reproducible, free-and-open-source pipeline (< 3 hrs on a laptop)
- A forward-simulation engine usable as a pre-investment diagnostic
- An interactive Power BI dashboard for non-technical SME managers

RECOMMENDATIONS

For SMEs: Diagnose Before You Invest

1. **Don't trust published % claims alone**

   Ask: at what data granularity, and was cost measured or assumed?

2. **Run the diagnostic on your own data first**

   The delivered simulation engine tells you if you face real stockout risk

3. **Invest where the diagnostic says risk is real**

   High-margin, fast-moving, or highly seasonal categories are the best candidates

4. **Deploy the dashboard regardless**

   Free, low-risk visibility — independent of which forecasting method feeds it

What This Study Does Not Claim

Limitations

- Single 56-day test window — a longer or differently-timed window could expose real stockouts
- LSTM evaluated at one architecture only — more elaborate sequence models untested
- Stockout cost modelled as linear — real-world costs (e.g. churn) may be non-linear

Future Work

- Re-test on retail benchmarks with more volatile demand (Rossmann, Favorita)
- Validate the boundary condition in a live SME deployment with genuine stockout exposure
- Extend the simulation engine to non-linear stockout costs and lead-time uncertainty

The Value of ML Forecasting Is Real — But Conditional, Not Automatic

This thesis delivers both the evidence for that conclusion and a practical, free, open-source tool that lets any organisation test which side of that condition it falls on — using its own data, before spending anything further.

The barrier this research removes is not financial. It is informational.

KEY SOURCES CITED

The Benchmark Figures and Frameworks Referenced Tonight

Highlighted subset of the 52 sources reviewed — full reference list is in the written thesis.

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10–25% / 5–15% benchmark figures

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Croston's method and SBA for intermittent demand

Thank You

Questions & Discussion

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